

Halvor Hafnor

hhafnor@gmail.com | linkedin.com/in/halvor-hafnor | github.com/halvis82 | halvorhafnor.vercel.app

SUMMARY

Embedded software and machine learning engineer building classification pipelines for real sensor data on constrained hardware, from collection through feature extraction to on-device inference. Two years at Aerendir Mobile, a shipped automotive product, a pending patent, and an M.S. at UC San Diego.

EXPERIENCE

Embedded ML Engineer

Aerendir Mobile

Jul. 2024 – Present

Mountain View, CA

- Developed embedded machine learning pipelines for classifying biometric motion signals from inertial sensors on **Linux**-based mobile devices.
- Built Android applications for real-time biometric data streaming and workflow control.
- Collected and classified the motion dataset behind the company's biometric authentication work: **400 participants**, each recorded holding the phone in three postures, sitting, standing and walking, from the accelerometer and gyroscope.
- Prototyped automotive driver monitoring from steering-wheel biometrics, shipped as **AerWheel**. Evaluated ST's on-sensor **Machine Learning Core**, then replaced it with a threshold classifier over hand-engineered vibration features running in the IMU's own firmware. Better than **99% accuracy** at under **0.5% false positives**. Subject of a pending patent naming me as an inventor.

Electrical Engineering Intern

ABB Electrification

Jun. 2021 – Sep. 2023

Skien, Norway

- Assembled and wired industrial electrical cabinets and control panels based on engineering schematics.
- Supported production and testing of custom electrical systems for industrial clients.

EDUCATION

University of California - San Diego - San Diego, CA

Computer Engineering, M.S.

Jan. 2026 – June 2027

Computer Engineering, B.S. Major GPA: 3.9

Sep. 2024 – June 2026

- **Provost Honors** every quarter enrolled, and **Tau Beta Pi**, the national engineering honor society.

De Anza College - Cupertino, CA

Computer Science & Engineering

GPA: 4.0

Academic and Technical Roles

Sep. 2022 – June 2024

- Math & Science Tutor and CS Teaching Assistant for calculus, physics, discrete math, C++, Java and Assembly.

TECHNICAL SKILLS

Languages: C++, C, Python, TypeScript, Java, Bash, ARM Assembly

Embedded: Embedded Linux, Android, QML/Qt, inertial sensing, on-sensor inference, ST IMU firmware, Docker

Signals & ML: Signal processing, feature extraction, classification, OpenCV, NumPy, SciPy, YOLOv8

Systems: Node, SQLite, WebSocket protocols

Agentic Development: Claude Code, MCP, spec-driven work breakdown, subagent orchestration, verification gates

PROJECTS

TabTerm, a macOS terminal in Chrome tabs

- A **Chrome extension** over a background **terminal daemon** that puts every shell session in a real Chrome tab, so tab groups, window management and keyboard shortcuts apply to terminals. Adds a command launcher, history search and project-scoped saved commands.

Laser targeting on an autonomous car, UC San Diego, ECE 148

- Real-time person tracking on a Raspberry Pi 5, in **Docker**. **YOLOv8** detection and **OAK-D** stereo depth, converted through the camera's field of view into pan and tilt angles so two servos hold a laser on the nearest person. Depth is the median of a patch, since one bad pixel throws the aim.